

## Er Diagram Exercises

### Exercise1

Draw an ER diagram for the following application from the manufacturing industry.

1. Each supplier has a unique name, and located in a city.
2. Each part has a unique number.
3. Each part has a color and a name.
4. A supplier can supply at least one part.
5. A part is supplied by one or more supplier.
6. A supplier is supplies part by a fixed quantity.

## Exercise 2

You have just been hired as a consultant for a big airplane manufacturer. They want you to completely redesign their database system. Talking with the people in the company, you get the following information.

- The database contains information about employees, factories and parts.
- Each employee has a social security number (SSN), name and salary. An employee is uniquely identified by his/her SSN. Also each employee has address composed of (city, state, and zip code). Each employee may have at least one duty.
- Each factory has an id, a name and a budget. The id uniquely identifies a factory.
- Each part has an id and a name. The id uniquely identifies a part.
- Each employee reports to at most one other employee.
- Each employee works in at least one factory. And of course a factory has many employees.
- Each part is manufactured in exactly one factory. A lot of different parts can be manufactured in a factory.

Draw the Entity-Relationship (ER) diagram for the above application.

### Exercise3

Notown Records has decided to store information about musicians who perform on its albums (as well as other company data) in a database.

- Each musician that records at Notown has unique SSN, a name, an address, and a phone number.

Poorly paid musicians often share the same address. Some musicians have more than one phone number.

- Each instrument used in songs recorded at Notown has a unique identification number, a name (e.g., guitar, synthesizer and flute) and a musical key (e.g., C, B-flat, E-flat).

- Each album recorded at Notown has a unique album identifier, a title, a copyright date, a format (e.g., CD or MC) and song count value. Song count value is derived by calculating how many songs exist in the album.

- Each song recorded at Notown has unique id , title and an author.

- Each musician may play several instruments, and a given instrument may be played by several musicians

- Each album has a number of songs on it, but each song appear on exactly one album.

- Each song is performed by one or more musicians, and a musician may perform a number of songs.

- Each album has exactly one musician who acts as its producer. A musician may produce several albums. Album produce has a start date and a cost.

#### Exercise4

UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system.

- Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date.
- Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address.
- Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). Each transportation event transports many items. These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute.

Please create an Entity Relationship diagram that captures this information about the UPS system. Be certain to indicate identifiers and cardinality constraints.

### Exercise5

Assume we have the following application that keeps track of the authors and their novels. In the design, we want to capture the following:

- We have a set of authors, each author has an ID (unique identifier), name, address composed of (city and state), and number of novels is calculated by the sum of novels.
- Each novel has a unique id, name, and price. A novel also can belong to more than one category.
- Novels are sold by bookstores. Each bookstore has a unique name, manager name and phone number. Each bookstore may have many addresses.
- A novel may be sold by one or more bookstore. A bookstore sells a lot of different novels.
- Each novel has a unique author.
- Each author may publish one or more novels. Publication has a publisher and a publication year.

**Design an ER diagram to capture the above requirements. State any assumptions you have that affects your design. Make sure cardinalities and primary keys are clear.**